

WIND TURBINE CONTROLLER VALIDATION REPORT

Technical Audit Certificate • Standard: IEC 61400-1 ed.4 / GL Guideline 2010 • Model: WT-5.0MW-126 (IEC Class IB)

OVERALL VERDICT: PASS

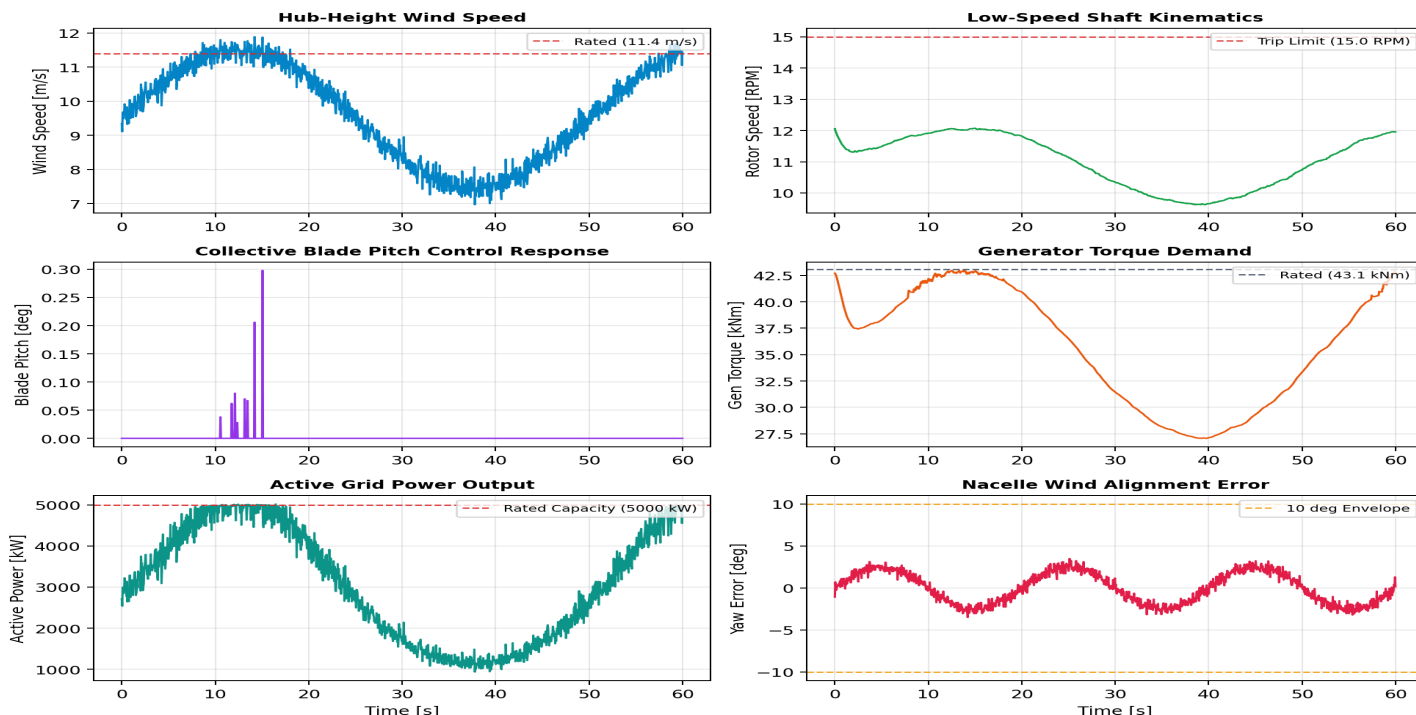
Validation Score: 100.0/100 | Tests Evaluated: 24 | Passed: 24 |
Warnings: 0 | Failed: 0

Run ID: run-83e8fb71
Dataset: normal_run.csv
Scenario: normal_run
Authority: Wind Energy Systems Division
Engineer: Lead Validation Engineer
Execution: 13.3 ms

1. Engineering Physical Key Performance Indicators

Metric Channel	Observed	Threshold Limit	Compliance
Max Rotor Speed	12.072 RPM	15.00 RPM	PASS
Max Generator Torque	42977.5 Nm	47,402.9 Nm	PASS
Peak Active Power	5020.1 kW	5,500.0 kW	PASS
Power Curve Deviation	100.0%	10.0%	FAIL
Pitch Actuator Latency	0.0 s	1.50 s	PASS
Peak Yaw Misalignment	3.5 deg	10.0 deg	PASS
Signal Completeness	100.0%	99.0%	PASS

2. Telemetry Signal Review



3. Structured Verification Evidence Log

Rule ID	Status	Metric (Actual vs Limit)	Engineering Diagnostic
signal_integrity.nan_dropout	PASS	nan_ratio_pct: 0.0 (Thr: 0.0)	Zero missing or NaN values detected across all telemetry channels.
signal_integrity.timestamp_duplicates	PASS	duplicate_timestamps: 0 (Thr: 0)	Timestamps are strictly unique.
signal_integrity.sampling_continuity	PASS	max_sampling_interval_sec: 0.0501 (Thr: 0.15)	Continuous sampling verified. Maximum dt = 0.0501s.
signal_integrity.wind_speed_range	PASS	out_of_bounds_wind_speed: 0 (Thr: 0)	Wind speed measurements within physical atmospheric boundary [0, 50] m/s.
signal_integrity.blade_pitch_range	PASS	invalid_pitch_samples: 0 (Thr: 0)	Blade pitch angles conform to mechanical actuator envelope [-2.0, 95.0] deg.
signal_integrity.sensor_freeze	PASS	frozen_sample_windows: 0 (Thr: 0)	Dynamic sensor variance confirmed (no frozen wind speed sensor detected).
rotor_speed.operating_range	PASS	max_rotor_speed_rpm: 12.072 (Thr: [6.9, 14.5])	Rotor speed strictly within nominal envelope [6.9, 14.5] RPM (min: 9.62, max: 12.07).
rotor_speed.rate_of_change	PASS	max_rotor_acceleration_rpm_s: 1.042 (Thr: 1.5)	Rotor acceleration dynamics smooth. Max d(omega)/dt = 1.042 RPM/s <= 1.5 RPM/s.
rotor_speed.gearbox_ratio	PASS	gearbox_ratio_error_pct: 0.0 (Thr: 2.5)	Drivetrain kinematic coupling verified. Gearbox ratio conforms to 97.0 (max error 0.00%).
overspeed.trip_boundary	PASS	peak_rotor_speed_rpm: 12.072 (Thr: 15.0)	Rotor speed securely below operational ceiling (12.072 RPM <= 14.50 RPM).
pitch_control.above_rated_response	PASS	mean_pitch_above_rated_deg: 0.0 (Thr: > 0.0 deg)	Wind speed remained below Region 3; pitch shedding not active.
pitch_control.actuator_rate_limit	PASS	max_pitch_rate_deg_per_sec: 5.96 (Thr: 8.0)	Pitch actuator rate within mechanical design envelope. Max rate = 5.96 deg/s <= 8.0 deg/s.

AUDIT SIGN-OFF: Certified that this automated assessment was executed with strict deterministic rules.
Controls Engineering Lead: *Lead Validation Engineer* | Date: 2026-10-07